

Arnesh Banerjee

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EDUCATION

Heritage Institute of Technology, Kolkata
Bachelor of Technology in Data Science

GPA: 8.876/10
Aug. 2023 – June. 2027

RESEARCH INTERESTS

Multi-Agent Reinforcement Learning (MARL) · RL Environments & Simulation · AI Safety & Alignment · Deep Learning · Applied ML · LLMs · AI

RESEARCH EXPERIENCE

IIT Kharagpur (Summer Internship 2026, Upcoming)

Starting 23rd May 2026

Advisor: Dr. Sourangshu Bhattacharya

On Site

- Selected for the **GRISHMA Summer Internship Program** at IIT Kharagpur via the official KRITI portal (kriti.iitkgp.ac.in); will work on India's first genomic foundational model (IgLM), a population-specific genomic language model built on the StripedHyena2 architecture.

Jadavpur University

November 2025 – Present

Advisor: Prof. Debotosh Bhattacharjee

Remote

- Achieved 98.10% Dice and 96.39% IoU on breast thermography tumor segmentation using a ResNet34 + ASPP + ViT decoder with SE attention, evaluated via 5-fold cross-validation.
- Conducted a rigorous ablation study (20 training runs) across U-Net, ResNet-only, and hybrid architectures, demonstrating that simpler models can match complex designs and highlighting generalization trade-offs in medical imaging.
- Built a fully automated cross-validation pipeline with checkpointing, experiment logging, and reproducible training, producing publication-ready comparative metrics for medical image analysis.

New Jersey Institute of Technology

June 2025 – July 2025, August 2025 - November 2025

Advisor: Dr. Arnob Ghosh

On Site, Virtual

- Constructed a cost-model training dataset for a modified Safe RLHF pipeline leading to improved results in the cost-model.
- Designed and tested comparative 800 datapoint safety benchmark against state-of-the-art LLMs (GPT, DeepSeek, Gemini, Qwen, Mistral).
- Demonstrated 50% higher safety compliance than ChatGPT 5 across benchmark scenarios.

Heritage Institute of Technology

October 2024 – March 2025

Advisor: Ms. Arpita Talukdar

On Site

- Developed a dual-stage ML framework for breast cancer diagnosis and recurrence prediction using WDBC and WPBC datasets, evaluating RF, SVM, Logistic Regression, MLP, and XGBoost with stratified 10-fold cross-validation.
- Implemented RFE and SFS feature selection with SMOTE and GridSearchCV tuning, improving WPBC accuracy to 93.67% (SVM + RFE) and WDBC to 97.77% (LogReg + RFE).
- Conducted comparative performance analysis and interpreted clinically relevant nuclear features, reporting stable cross-validated results with confidence intervals.

PUBLICATIONS

- Ayushi Bhattacharjee, **Arnesh Banerjee**, Arpita Talukdar. 2026. Recursive and Wrapper-Based Feature Selection for Breast Cancer Diagnosis and Prognosis. In the proceedings of the 4th Analytics Global Conference (AGC 2026), March 2026. **Oral**.

PRE-PRINTS

- Kartik Pandit, Sourav Ganguly, **Arnesh Banerjee**, Shaahin Angizi, Arnob Ghosh. 2025. Certifiable Safe RLHF: Fixed-Penalty Constraint Optimization for Safer Language Models. arXiv preprint [arXiv:2510.03520](https://arxiv.org/abs/2510.03520).
- Arnesh Banerjee**, Debotosh Bhattacharjee. Multiscale Transformer-Enhanced U-Net with Level-Set Regularization for Breast Thermography Segmentation. *Advisor: Prof. Debotosh Bhattacharjee*

INDEPENDANT & ONGOING RESEARCH

- **Arnesh Banerjee**. Co-evolutionary Multi Agent RL for Autonomous Drones. In collaboration with the AI for Defence Lab, ULiège, Belgium.
- Kartik Pandit, Sourav Ganguly, **Arnesh Banerjee**, Ayushi Bhattacharjee, Avirup Chakraborty, Arnob Ghosh. Analyzing Historical Revisionism in LLMs in the Context of Indian History. *Advisor: Dr. Arnob Ghosh.*
- **Arnesh Banerjee**, Ayushi Bhattacharjee, Subhajit Datta. Understanding the Limitations of LLMs in Mathematical Reasoning. *Advisor: Prof. Subhajit Datta.* Part of B.Tech degree coursework.

ACHIEVEMENTS

- **Department Third:** 4th Semester. SGPA: 9.46. B.Tech CSE(DS), Heritage Institute of Technology, Kolkata
- **Institutional Innovation Council(IIC):** Selected one of 10 IIC members across all years to represent the Department of CSE(DS), HIT, Kolkata
- **Paper Presenter:** Presenting Author for paper at AGC 2026 on Applied ML in Medical domain.
- **WBJEE Qualified:** Top 5.3% in the state on WBJEE, 2023

TECHNICAL SKILLS

Languages: Python, R, C, Java, SQL, LaTeX

Machine Learning: PyTorch, Scikit-Learn, Keras, H2O, CatBoost, XGBoost, Reinforcement Learning, Deep Q-Networks (DQN), Double DQN, PPO, SAC

Simulation & RL Environments: Gymnasium, OpenAI Gym, Stable-Baselines3, JAX

Data & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Plotly

Frameworks & Tools: Flask, FastAPI, Shiny, Docker, Git

Relevant Courses: Operating Systems, Database Management System, Data Structures, Algorithms, Machine Learning, Data Mining, Data Warehousing